Transformer's Revenge: Stress-Testing minGRU's Claims Against Attention on Algorithmic Reasoning Tasks

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Abstract

Feng et al. [2024] recently asked "Were RNNs All We Needed?", showing that a minimal gated recurrent unit (minGRU) matches Transformer performance on standard language modelling benchmarks with $O(T)$ complexity. We stress-test this claim by moving beyond language modelling into tasks requiring explicit algorithmic reasoning (verbatim sequence copying and induction head pattern completion) where architectural limitations are exposed rather than masked by statistical regularity. In a controlled setting ($\sim$750–800K parameter target, identical optimiser and schedule), we compare minGRU against a Transformer with Rotary Position Embeddings (RoPE) and Root Mean Square Normalisation (RMSNorm), and a causal gated MLP (gMLP) architecture we adapt as a third paradigm bridging fixed-parameter spatial mixing and learned attention. Our 27-experiment grid (3 models $\times$ 3 tasks $\times$ 3 sequence lengths) demonstrates that minGRU's input-only gating fundamentally prevents content-based retrieval at *any* sequence length, while gMLP succeeds within its fixed spatial window but fails beyond it. The Transformer remains the only architecture capable of unbounded algorithmic reasoning within training budget. We additionally observe a sharp phase transition in Transformer copy learning, consistent with recent work on circuit formation.

1 Introduction

Feng et al. [2024] make a provocative claim: by stripping the GRU to its minimal form and enabling parallel training via log-space scans, the resulting minGRU matches Transformer performance on language modelling while offering $O(T)$ linear complexity. Their experiments (character-level Shakespeare, a selective copying task, and reinforcement learning locomotion) support this claim within those domains. But language modelling primarily tests statistical pattern learning from local context, a setting where even simple n-gram models perform surprisingly well. The question we ask is: *does the claim hold when we move beyond statistical modelling into tasks requiring explicit algorithmic reasoning?*

We engage directly with the minGRU paper by reproducing their architecture in a controlled setting and then subjecting it to tasks specifically designed to test capabilities that the architectural analysis suggests should be impossible. Our experimental design is a stress test: we already suspect where minGRU will fail based on its mathematical structure, but we verify this empirically and precisely characterise the failure boundaries.

To contextualise minGRU's capabilities, we compare against two architectures:

- **A nanoGPT-based Transformer** enhanced with key LLaMA-recipe modifications [Touvron et al., 2023]: RoPE [Su et al., 2024] for relative position encoding and RMSNorm [Zhang and Sennrich, 2019] for stable pre-norm residuals, with all biases removed. We use GELU rather than SwiGLU for the FFN activation, as SwiGLU's benefits are less pronounced at small model scales and we observed training instability at this parameter budget in prior coursework (COMP8650 AML). This represents the attention-based upper bound.
- **A causal gMLP** [Liu et al., 2021]: Our contribution adapting the originally bidirectional gMLP for autoregressive generation as a third paradigm. The gMLP uses fixed spatial weights rather than data-dependent attention, placing it architecturally *between* the minGRU (no cross-position interaction beyond recurrence) and the Transformer (full data-dependent cross-position interaction). This allows us to disentangle *whether* a model can mix across positions from *how* it mixes.

All models use 4 layers; the theoretical and empirical justification for this depth choice is discussed in Section 3.2.

We test on three tasks of increasing algorithmic difficulty: character-level language modelling (TinyShakespeare), verbatim sequence copying, and induction head pattern completion, each at three sequence lengths chosen to cross architectural boundaries. This yields a $3 \times 3 \times 3 = 27$ experiment grid from which we derive predictions *a priori* and then verify experimentally.

2 Background and Architectural Analysis

2.1 Transformer with LLaMA-Inspired Modifications

Our Transformer builds on the nanoGPT codebase, enhanced with key modifications from the LLaMA recipe [Touvron et al., 2023]: pre-norm residual connections using RMSNorm [Zhang and Sennrich, 2019], Rotary Position Embeddings (RoPE) [Su et al., 2024], and removal of all bias terms. The core mechanism is multi-head causal self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V \tag{1}$$

where M is a causal mask ($M_{ij} = -\infty$ for $j > i$). RoPE rotates query and key vectors by position-dependent angles, enabling attention scores to depend only on relative distance $m - n$ without learned absolute embeddings.

Key property: Every token attends to *every* preceding token with a learned, *content-dependent* weight. The attention matrix provides unbounded context access; the only constraint is $O(T^2)$ compute cost.

2.2 minGRU: The Claim Under Test

The minGRU [Feng et al., 2024] strips the classical GRU to its minimal form, removing the reset gate entirely and enabling parallel training via log-space cumulative sums:

$$g_t = \sigma(W_g x_t + b_g) \tag{2}$$

$$\tilde{h}_t = W_h x_t + b_h \tag{3}$$

$$h_t = (1 - g_t) \odot h_{t-1} + g_t \odot \tilde{h}_t \tag{4}$$

The parallel form computes all hidden states simultaneously:

$$\log h_t = \text{logcumsumexp}\left(\sum_{k=1}^{t} \log(1 - g_k),\ \log g_t + \log \tilde{h}_t\right) \tag{5}$$

This achieves $O(T)$ complexity and enables efficient GPU utilisation through parallel scan operations. Feng et al. [2024] demonstrate competitive perplexity on language modelling benchmarks, motivating their claim.

Critical limitation: The gate g_t in Eq. 2 depends *only on the current input* x_t, not on the hidden state h_{t-1}, not on any future query about what to retrieve. The model must decide at write-time what to store, without knowledge of what will be needed later. For tasks requiring content-based retrieval ("find the token that appeared after A"), this is architecturally insufficient: the gate cannot implement a selective read operation conditioned on a query. This is the specific weakness we target in our stress test.

2.3 Causal gMLP: A Third Paradigm (Our Contribution)

We adapt the originally *bidirectional* gMLP architecture [Liu et al., 2021] for autoregressive generation as a third comparison point. The original gMLP was proposed for classification tasks using full $T \times T$ spatial projections; we modify it with causal masking and a windowed SGU for efficient generation. The gMLP replaces attention with a Spatial Gating Unit (SGU) that applies a learned linear projection across the sequence dimension:

$$U, V = \text{split}(\text{Linear}(x)) \tag{6}$$

$$V' = (W_s V + b_s) \quad \text{where } W_s \in \mathbb{R}^{T \times T} \tag{7}$$

$$\text{SGU}(x) = U \odot V' \tag{8}$$

For causal modelling, W_s is masked to be lower-triangular. We restrict W_s to a banded structure with window size $w = 256$:

$$(W_s)_{ij} = \begin{cases} w_{ij} & \text{if } 0 \leq i - j < w \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

The gMLP occupies a unique position in our comparison: unlike minGRU, it *does* perform explicit cross-position information mixing. Unlike the Transformer, this mixing uses *fixed* parameters rather than content-dependent queries. This disentangles two questions: (1) can the model route information across positions at all? (2) can it do so in a content-dependent manner?

Critical limitation: The spatial weights W_s are fixed parameters: token t aggregates from $[t - w + 1, \ldots, t]$ with weights independent of content. Beyond window w, information is inaccessible in a single layer. While L layers give a theoretical receptive field of $L \times w$, precise token-level retrieval beyond one window is impractical.

2.4 A Priori Predictions

From the architectural analysis above, we derive three predictions before running experiments:

1. **Transformer**: should succeed on all tasks. Attention provides unbounded, content-dependent access to all prior tokens; no architectural barrier exists for copy or induction at any length.

2. **minGRU**: will fail on all algorithmic tasks regardless of sequence length. Input-only gating (Eq. 2) cannot implement content-based retrieval; this is a structural impossibility, not a capacity issue.

3. **gMLP**: will succeed on algorithmic tasks where the required retrieval distance falls within its 256-token window, and fail beyond it. The boundary should be sharp, not gradual.

All three models should achieve comparable performance on Shakespeare, since language modelling relies on local statistical patterns, precisely the domain where Feng et al. [2024] demonstrated parity. The algorithmic tasks, by contrast, require *exact* content-based retrieval across arbitrary distances, which directly tests the architectural limitations identified above.

3 Experimental Setup

3.1 Tasks and Data

TinyShakespeare. Character-level language modelling on the 1.1M-character Shakespeare corpus (65-character vocabulary). We use a 90/10 train/validation split. This directly mirrors the type of benchmark on which Feng et al. [2024] demonstrated minGRU's competitiveness. Metric: validation perplexity.

Long-Range Copy. Sequences of the form `[content][separator][content]`, where the model must reproduce the first half verbatim after the separator. Content is drawn from a 26-character alphabet. Short: T=128 (64 tokens to memorise), Medium: T=528 (256 tokens), Long: T=2048 (1024 tokens). This task requires pure storage and retrieval; no statistical shortcut exists. Metric: recall perplexity (computed only on the recalled portion).

Induction Heads. Sequences containing repeated 5-character patterns: after observing a pattern $[P_0P_1P_2P_3P_4]$ earlier in the sequence, the model must predict the remaining characters when the pattern prefix reappears. Generated at $T \in \{128, 256, 2048\}$ with one induction pattern per sequence. This tests content-based retrieval: the model must use the current context as a *query* to find what followed that context previously. Metric: masked accuracy on induction positions only.

3.2 Model Configurations

All models use 4 layers with hidden dimension 128, targeting ∼750–800K parameters:

Table 1: Model configurations. All share 4 layers and d=128.

	Transformer	**gMLP**	**minGRU**
Parameters	∼795K	∼508–933K†	∼737K
Layers	4	4	4
Hidden dim	128	128	128
Attention/Window	4 heads	w=256	–
FFN	GELU, $4 \times d$	GELU, d_{ffn}=512	GELU, $4.5 \times d$
Normalisation	RMSNorm	LayerNorm	RMSNorm
Position encoding	RoPE	Learned	Implicit (recurrence)
Additional	bias=False	–	Conv1d (k=4)

†gMLP parameter count varies with sequence length due to the $T \times T$ spatial weight matrix within the SGU.

The depth of 4 layers is motivated by theoretical results showing exponential expressiveness gains from depth [Telgarsky, 2016]: deeper networks can represent functions requiring exponentially more parameters in shallower architectures. The hierarchical composition enabled by depth [Bengio and Delalleau, 2011, Montufar et al., 2014] is particularly important for our algorithmic tasks, where multi-step reasoning (identify pattern, locate match, retrieve next token) benefits from layered processing. Our depth choice aligns with Feng et al.'s finding that 3 layers are required for even simple memory tasks (1-layer achieves only 37% on selective copy vs. 99.5% at 3 layers).

3.3 Training Protocol

All 27 experiments share an identical training protocol to ensure fair comparison:

- Optimiser: AdamW (β_1=0.9, β_2=0.95, weight decay 0.1)
- Learning rate: cosine decay from 3×10^{-4} to 3×10^{-5}
- Warmup: 200 steps (linear)
- Dropout: 0.05
- Gradient clipping: max norm 1.0
- Batch size: 64 (reduced to 32 for gMLP at T=2048 due to $O(T^2)$ memory)
- Steps: 5,000 for Shakespeare and Copy; 20,000 for Induction
- Hardware: NVIDIA A100 GPU (Google Colab)

3.4 Implementation

The minGRU implementation directly follows Feng et al. [2024], reproducing their Appendix C.2 language modelling recipe: each block consists of a causal Conv1d (k=4), the minGRU cell, and an MLP, all with pre-norm RMSNorm and residual connections. The Transformer is built from nanoGPT with LLaMA-inspired modifications (RoPE, RMSNorm, no bias), reflecting a subset of the recipe validated at scale [Touvron et al., 2023]. The causal gMLP adapts Liu et al. [2021] with a banded causal SGU (window w=256), as described in Section 2.3. Each model has an independent training script with shared interfaces for task specification. Data generation for synthetic tasks uses identical random processes across models. Source code is available at `https://github.com/Lord-V15/DL-RNNs-Revenge`.

4 Results

4.1 Overview

Figure 1 presents the complete 27-experiment summary. The central finding is immediately visible: all models achieve comparable language modelling performance (confirming Feng et al.'s claim in that domain), but algorithmic tasks expose sharp architectural boundaries that the language modelling evaluation entirely misses.

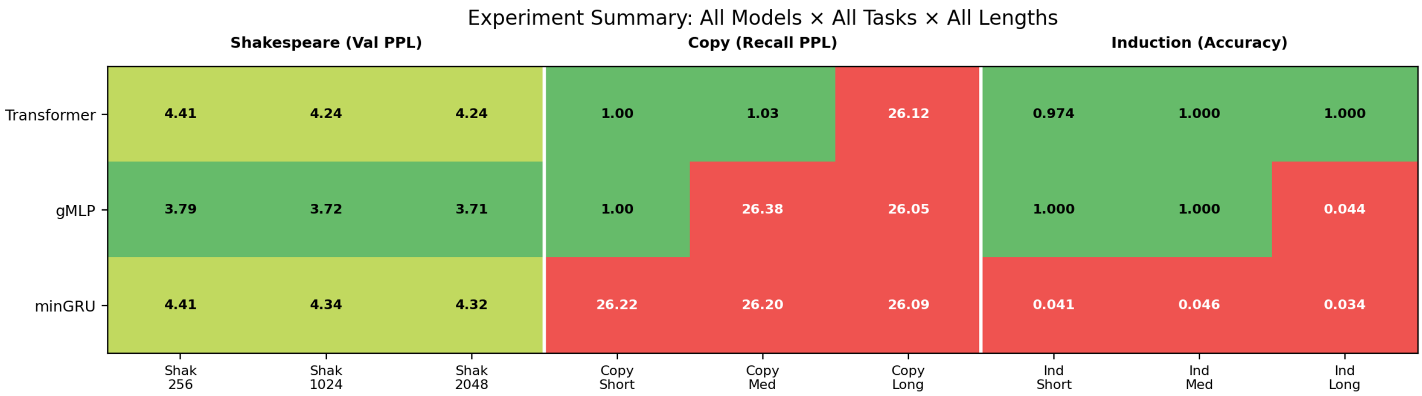
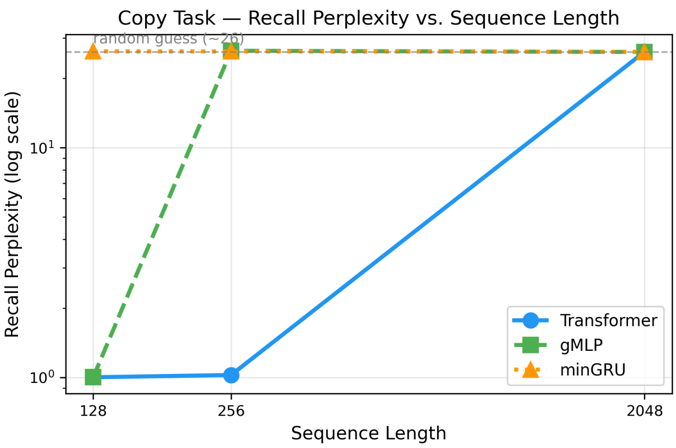


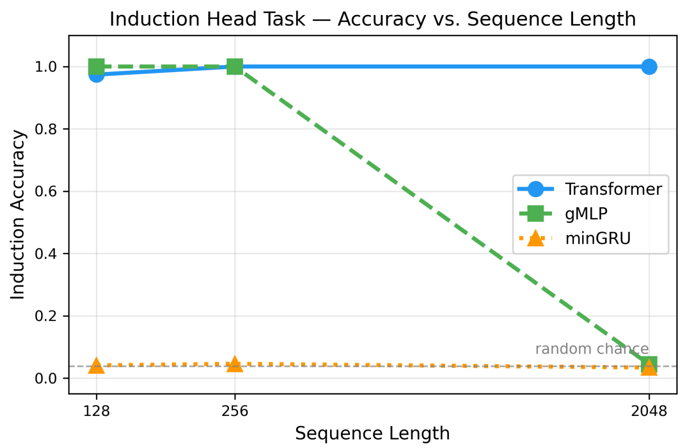
Figure 1: Summary of all 27 experiments. Shakespeare: validation perplexity (lower = better). Copy: recall perplexity (1.0 = perfect, ∼26 = random). Induction: masked accuracy (1.0 = perfect, ∼0.04 = random). Colour: green = pass, yellow = partial, red = fail.

4.2 Stress Test: Algorithmic Tasks

Figure 2 shows performance as a function of sequence length for both algorithmic tasks, the core of our stress test.



(a) Copy task: recall perplexity (log scale).



(b) Induction task: induction accuracy (final character completion).

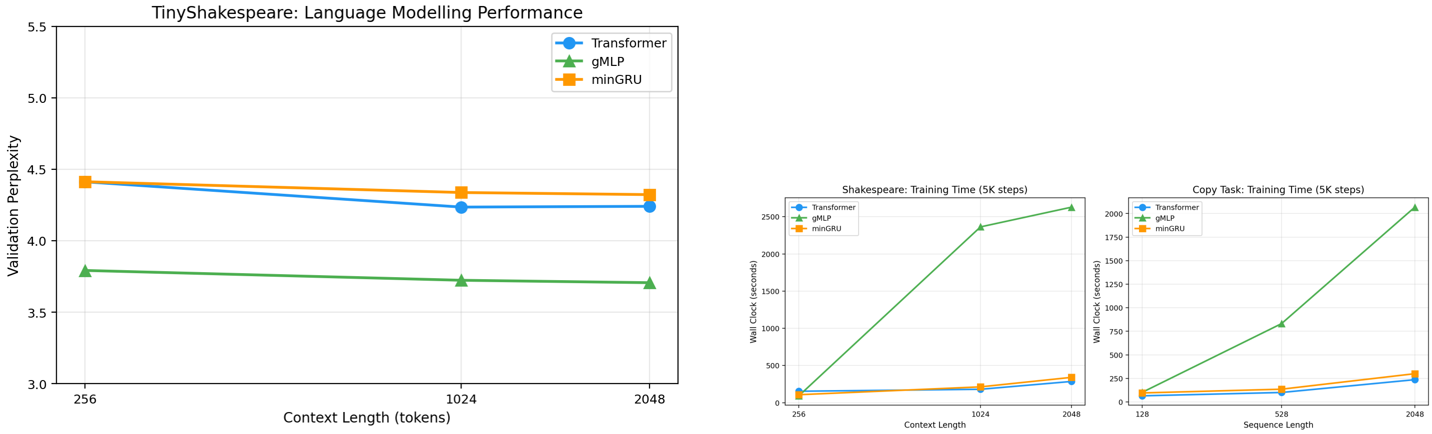
Figure 2: Algorithmic task performance vs. sequence length. Each model's failure point maps precisely to its architectural limitation: minGRU fails at all lengths (no content-based retrieval), gMLP fails beyond its 256-token window, Transformer succeeds except where training budget is insufficient.

Copy task (Figure 2a): The Transformer achieves near-perfect recall (PPL ≈1.0) at short and medium lengths but fails at T=2048 (PPL = 26.12, equivalent to random). The gMLP succeeds at T=128 (within window) but fails at T=528 where the recall span exceeds its 256-token boundary. The minGRU fails at *all* lengths (PPL ≈26), confirming that the limitation is architectural, not capacity-related.

Induction task (Figure 2b): The Transformer achieves ≥0.97 accuracy at all lengths. The gMLP matches this at T=128 and T=256 (within window) but collapses to random chance (0.044) at T=2048. The minGRU achieves only random-chance accuracy (≈0.04) regardless of sequence length; it cannot implement the "query by content" operation that induction requires.

Key result: Feng et al.'s claim that minGRU is competitive with Transformers holds strictly for statistical language modelling but *categorically fails* for algorithmic reasoning tasks. The failure is not due to insufficient capacity or training; it is a fundamental architectural limitation of input-only gating.

4.3 Confirming the Claim: Language Modelling



(a) TinyShakespeare validation perplexity.

(b) Wall-clock training time vs. sequence length.

Figure 3: (a) All models achieve comparable language modelling perplexity, confirming Feng et al.'s claim in that domain. (b) Wall-clock training time. Transformer benefits from optimised GPU kernels; minGRU scales most gently; gMLP is slowest at long sequences.

On TinyShakespeare (Figure 3a), all three architectures achieve similar perplexity (3.7–4.4), confirming Feng et al.'s central result. In their own experiments, the Transformer achieves validation loss 1.547 versus minGRU's 1.548, near-identical parity that our results mirror. The gMLP achieves the lowest perplexity (3.71), consistent with natural language being predominantly local, though single-seed variance prevents us from drawing strong conclusions from this gap. These results validate that the algorithmic task failures are not artefacts of implementation quality or insufficient model capacity.

Figure 3b shows wall-clock training time. The Transformer benefits from highly optimised GPU attention implementations in PyTorch, becoming fastest at long sequences. minGRU's $O(T)$ parallel scan shows the gentlest growth with sequence length, suggesting it would overtake the Transformer at longer sequences where $O(T^2)$ attention becomes dominant. The gMLP is slowest at long sequences due to its unoptimised $T \times T$ spatial matrix, requiring batch size reduction to 32 at T=2048 to fit in memory.

4.4 Phase Transition: Evidence of Circuit Formation

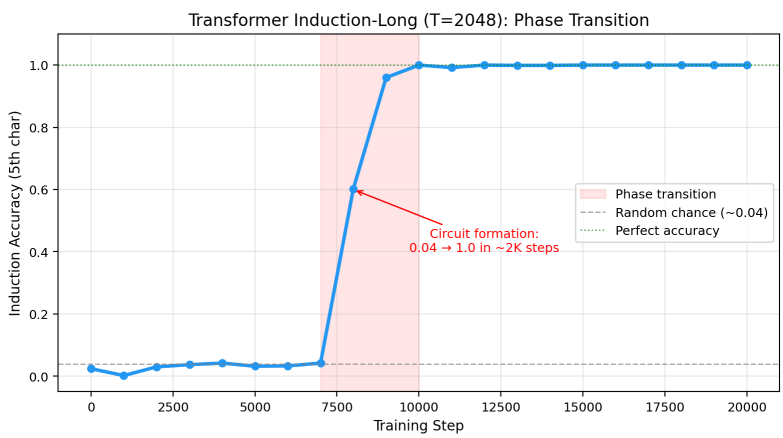


Figure 4: Training dynamics for Transformer on induction-long (T=2048). The model shows minimal improvement for approximately 7,000–8,000 steps, then abruptly jumps to perfect accuracy between steps 8,000 and 10,000.

Figure 4 reveals a striking training dynamic: the Transformer's induction-long accuracy shows minimal improvement through ~7,000 steps, then undergoes a rapid phase transition to perfect accuracy (1.0) between steps 8,000 and 10,000. This is consistent with "grokking" [Power et al., 2022] and the circuit formation hypothesis [Olsson et al., 2022]: the attention heads abruptly snap into an induction circuit configuration. We hypothesise that the Transformer's failure at T=2048 copy may similarly reflect insufficient training (5K steps) for a phase transition to occur, rather than an architectural limitation, though we lack direct evidence of such a transition beginning in the copy-long training curve.

5 Discussion

5.1 The Boundaries of "Were RNNs All We Needed?"

Our results precisely delineate where Feng et al.'s claim holds and where it breaks:

- **Claim holds**: For language modelling tasks dominated by local statistical patterns, minGRU is competitive with Transformers while being faster to train. Our Shakespeare results confirm this.
- **Claim breaks**: For any task requiring *content-based retrieval* (using the current input as a query to selectively access past information), minGRU categorically fails. This is not a training or scale issue; it is a mathematical consequence of Eq. 2.

The distinction maps to a concrete architectural criterion: does the task require *data-dependent* information routing? If the answer is yes, input-only gating is insufficient.

It is important to distinguish our findings from Feng et al.'s own "selective copying" experiment (adapted from Gu and Dao 2023), where minGRU achieves 99.5% accuracy. Their task is *content-aware filtering*: certain tokens are marked as important at encoding time, and the model need only retain those marked tokens while ignoring distractors. This succeeds because the gate g_t can learn to open for marked tokens; it is a *write-time* decision. Our tasks are fundamentally harder: in verbatim copy, *all* tokens must be stored; in induction heads, retrieval is conditioned on a future query (a *read-time* decision). This is precisely the boundary that input-only gating cannot cross, as noted by Merrill and Sabharwal [2024] in their theoretical analysis of state-space expressiveness without state-dependent gating.

5.2 Predictions vs. Reality

Table 2 compares our a priori predictions against observed outcomes:

Table 2: Predicted vs. observed outcomes. All architectural predictions confirmed.

Experiment	Predicted	Observed	Match
minGRU Copy (all lengths)	×	× (PPL ≈26)	✓
minGRU Induction (all)	×	× (Acc ≈0.04)	✓
gMLP Copy-Short	✓	✓ (PPL=1.003)	✓
gMLP Copy-Medium	×	× (PPL=26.4)	✓
gMLP Induction-Long	×	× (Acc=0.044)	✓
Transformer Copy-Long	✓	× (PPL=26.1)	×*
All Shakespeare	✓	✓	✓

*Training budget limitation, not architectural (see Section 4.4).

All predictions were confirmed except one: the Transformer failed on copy-long despite having no architectural barrier. We hypothesise (Section 4.4) that this reflects a training budget limitation — the attention mechanism may require more than 5K steps to discover the circuit at T=2048 — rather than an architectural one. This interpretation, while not directly evidenced in our copy-long training curve, would reinforce rather than undermine the core finding: only attention *can* solve these tasks, even if it sometimes needs more time to do so.

One might argue our results reflect confirmation bias, that we chose tasks designed to expose failures. We contend the opposite: if the claim is that RNNs are "all we needed," then the architecture should succeed on tasks the Transformer handles routinely. Induction heads and copy are not exotic benchmarks; they are fundamental capabilities that emerge naturally in trained Transformers [Olsson et al., 2022]. A model that claims to replace Transformers must handle Transformer-native tasks.

5.3 Implications: The gMLP Diagnostic and Architecture Selection

The gMLP's role in our study is diagnostic: it answers whether cross-position mixing is sufficient, or must be content-dependent. Within its window, gMLP achieves perfect algorithmic performance: fixed spatial weights *can* learn arbitrary routing over a bounded range. Beyond it, performance collapses discontinuously (not gradually), confirming a *hard* boundary. This positions gMLP as architecturally intermediate: it has cross-position mixing (unlike minGRU) but not data-dependent mixing (unlike Transformer), confirming that the critical ingredient for unbounded algorithmic reasoning is content-dependent attention.

These findings yield actionable guidance: (1) for statistical/local tasks, minGRU is viable and efficient; (2) for bounded retrieval with known maximum distance, gMLP with window $\geq$ that distance suffices; (3) for unbounded or content-addressed retrieval, attention remains necessary.

6 Limitations and Future Work

Parameter matching. The gMLP's spatial weight matrix scales with T, making exact parameter matching across lengths impossible. At T=2048, gMLP has ∼933K parameters vs. ∼795K for others; this *favours* gMLP, making its long-range failures more striking.

Normalisation and position encoding. Models use different normalisation (RMSNorm vs. LayerNorm) and position encoding (RoPE, learned, implicit). We follow each architecture's original specification rather than artificially unifying components. These differences are unlikely to explain pass/fail boundaries: both norms are mathematically similar at our scale, and position encoding differences would favour gMLP (exact learned positions) over the Transformer on copy tasks.

Single seed. All experiments use seed 42. While results are qualitatively unambiguous (pass/fail), multiple seeds would strengthen quantitative claims.

Batch size. gMLP at T=2048 uses batch size 32 (vs. 64) due to memory constraints, resulting in half the gradient samples per step.

Future directions:

- Extend Transformer copy-long training beyond 5K steps to verify whether the phase transition eventually occurs (our analysis predicts it should).
- Compare with Mamba [Gu and Dao, 2023], which has *state-dependent* gating; we predict it would overcome minGRU's limitation while retaining $O(T)$ complexity.
- Test minLSTM [Feng et al., 2024], which uses separate forget and input gates (f_t, i_t) providing a theoretically richer gating mechanism than minGRU's single gate. We chose minGRU over minLSTM for this study due to its superior training stability (Feng et al. report variance up to ±5.8% for minLSTM vs. ±0.2% for minGRU); however, the separate gates may enable partial content-based retrieval that the single-gate minGRU cannot.
- Multi-seed experiments: our results are qualitatively unambiguous (pass/fail), but statistical confidence intervals would strengthen quantitative claims. Time budget constraints limited us to seed 42 across all 27 experiments.
- Attention head visualisation to confirm the hypothesised copy/induction circuit structure.
- Test hybrid architectures (gMLP + sparse global attention) to find the minimal attention needed for long-range capability.

7 Conclusion

Our 27-experiment stress test shows that Feng et al.'s claim holds for statistical language modelling but categorically fails for algorithmic reasoning. The answer to "were RNNs all we needed?" is "for which tasks?", and the boundary is precisely whether the task requires data-dependent information routing. When it does, content-dependent attention remains necessary; no efficient alternative tested here can substitute.

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Declaration

Project Topic

This report addresses Topic 3 (Sequence Modelling: RNN vs Attention) from the COMP6242 Group Project Ideas.

Individual Contributions

- **Mayukh Das**, ID: u7965027: gMLP paper selection, Transformer implementation, experiment design, architecture comparison framework, implementation ideation, bug fixes, version control and repository management, report writing, presentation - recording, scripting and video editing, paper engagement (Feng et al., gMLP)
- **Vibhansh Gupta**, ID: u7861976: Feng et al. paper selection, causal gMLP implementation, version control and repository management, implementation ideation, presentation recording, paper engagement (Feng et al., gMLP)
- **Daniel Vaz**, ID: u7990536: minGRU implementation, implementation ideation, presentation recording, paper engagement (Feng et al., gMLP)
- **Adam B Clark**, ID: u7437561: paper engagement (gMLP)

Use of Generative AI

Claude (Anthropic) was used to assist with: training script scaffolding, debugging runtime errors, plotting code, and LaTeX formatting of this report. The research idea, paper selection, experimental design, architectural analysis, hypothesis formation, and interpretation of results are entirely the team's original work, developed through engagement with COMP6242 course material and prior coursework. All AI-generated code and text were reviewed, verified, and modified by team members before use.